MD HASANUR RAHMAN

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EDUCATION

PhD in Electrical and Computer Engineering *Expected Spring 2027*
University of Florida *2025–Present*
University of Iowa *2021–2025*
(Transferred to Florida Following Advisor's Move in 2025)
Advisor: Guanpeng Li

MS in Computer Science *2024*
University of Iowa *2021–2024*

BS in Computer Science and Engineering *2019*
Bangladesh University of Engineering and Technology *2015–2019*

RESEARCH INTERESTS

High-Performance Computing, System Reliability and Fault Tolerance, Program Analysis, Data Reduction, Reliable and Efficient AI.

SELECTED RESEARCH CONTRIBUTIONS

- **Execution-aware reliability characterization.** Developed execution-aware methods to uncover silent data corruption vulnerabilities across CPU, GPU, and intelligent workloads, enabling reliability evaluation beyond traditional static or average-case testing.
- **Adaptive fault protection and error propagation.** Developed lightweight methods for context-aware fault protection and temporal error-propagation analysis, connecting low-level hardware faults to application-visible outcomes for targeted validation and protection.
- **Data-aware compression and resource management.** Developed models and decision frameworks that use data characteristics and downstream requirements to guide lossy compression, replacing trial-and-error configuration with resource-aware decisions.

PUBLICATIONS

Peer-Reviewed Conference Publications (10)

- **Full-Coverage Variable-Precision Prompt KV Cache Compression for Long-Context LLM Inference**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Conference on Big Data (BigData), 2026. To appear.
  Acceptance rate: 19.4%
- **Temporal Modeling of Soft Error Propagation Using Dynamic Bayesian Networks**
  Ahmer Jamil, Md Hasanur Rahman, Bingzhe Li, Xiaoyi Lu, Guanpeng Li
  IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2026. To appear.
  Acceptance rate: 19.2%

- **Deploying Lightweight Input-Aware Selective Instruction Duplication in HPC Applications**
  Md Hasanur Rahman, Guanpeng Li
  ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2025.
  Acceptance rate: 21.2%
- **Modeling Rate-Distortion for Endpoint-Aware Lossy Compression in Scientific Data Transfer**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Performance Computing and Communications Conference (IPCCC), 2025.
  Acceptance rate: 32.1%
- **Druto: Upper-Bounding Silent Data Corruption Vulnerability in GPU Applications**
  Md Hasanur Rahman, Sheng Di, Shengjian Guo, Xiaoyi Lu, Guanpeng Li, Franck Cappello
  IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024.
  Acceptance rate: 26.1%
- **A Generic and Efficient Framework for Estimating Lossy Compressibility of Scientific Data**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Conference on Massive Storage Systems and Technology (MSST), 2024.
- **Significantly Improving Fixed-Ratio Compression Framework for Resource-limited Applications**
  Tri Nguyen, Md Hasanur Rahman, Sheng Di, Michela Becchi
  International Conference on Parallel Processing (ICPP), 2024.
  Acceptance rate: 29.0%
- **A Feature-Driven Fixed-Ratio Lossy Compression Framework for Real-World Scientific Datasets**
  Md Hasanur Rahman, Sheng Di, Kai Zhao, Robert Underwood, Guanpeng Li, Franck Cappello
  IEEE International Conference on Data Engineering (ICDE), 2023.
  Acceptance rate: 30.6%
- **Characterizing Deep Learning Neural Network Failures between Algorithmic Inaccuracy and Transient Hardware Faults**
  Sabuj Laskar, Md Hasanur Rahman, Bohan Zhang, Guanpeng Li
  IEEE Pacific Rim International Symposium on Dependable Computing (PRDC), 2022.
  Acceptance rate: 36.0%
- **Peppa-X: Finding Program Test Inputs to Bound Silent Data Corruption Vulnerability in HPC Applications**
  Md Hasanur Rahman, Aabid Shamji, Shengjian Guo, Guanpeng Li
  ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
  Acceptance rate: 23.6%

Peer-Reviewed Journal Publications (2)

- **A Survey on Error-Bounded Lossy Compression for Scientific Datasets**
  Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, Yafan Huang, Jiajun Huang, Xiaodong Yu, Congrong Ren, Hanqi Guo, Grant Wilkins, Dingwen Tao, Jiannan Tian, Sian Jin, Zizhe Jian, Daoce Wang, Md Hasanur Rahman, Boyuan Zhang, Shihui Song, Jon Calhoun, Guanpeng Li, Kazutomo Yoshii, Khalid Alharthi, Franck Cappello
  ACM Computing Surveys, 2025.
- **Investigating the Impact of Transient Hardware Faults on Deep Learning Neural Network Inference**
  Md Hasanur Rahman, Sabuj Laskar, Guanpeng Li
  Software Testing, Verification and Reliability (STVR), 2024.

Workshop Publications (3)

- **Characterizing Spatial Data Traits for Modeling Generic Lossy Rate-Distortion Quality**
  Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
  IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPS-W), 2025.
- **LibPressio-Predict: Flexible and Fast Infrastructure for Inferring Compression Performance**
  Robert R. Underwood, Sheng Di, Sian Jin, Md Hasanur Rahman, Arham Khan, Franck Cappello
  ACM International Workshop on Data Reduction for Big Scientific Data (DRBSD), in conjunction with SC, 2023.
- **TensorFI+: A Scalable Fault Injection Framework for Modern Deep Learning Neural Networks**
  Sabuj Laskar, Md Hasanur Rahman, Guanpeng Li
  IEEE International Workshop on Resiliency, Security, Defences and Attacks (ISSRE-W), 2022.

RESEARCH EXPERIENCE

University of Florida / University of Iowa *2021–Present*
Doctoral Research on Dependable and Data-Efficient Computing *USA*

· Conduct research on reliability, fault characterization and protection, program analysis, and data reduction for large-scale computing systems.

Argonne National Laboratory *2021–2026*
Graduate Remote Research Intern *USA*

· Conduct research on scientific data compression, compressibility modeling, and resource-aware data movement in collaboration with researchers at Argonne and partner institutions.

TEACHING EXPERIENCE

Programming for Electrical Engineers II *Fall 2026*
Teaching Assistant, University of Florida

Discrete Structures *Fall 2021*
Teaching Assistant, University of Iowa

RESEARCH MENTORING

- **Ahmer Jamil, PhD student (2025–2026):** Mentored research on modeling error propagation and fault behavior in HPC systems; work recently accepted at SC 2026.
- **Sabuj Laskar, research intern (2022–2023):** Mentored research on transient hardware faults in deep neural network inference; work resulted in publications at PRDC 2022 and ISSRE-W 2022.

INDUSTRY EXPERIENCE

Samsung Research *2019–2021*
Software Engineer, Bangladesh

HONORS AND AWARDS

- SC Travel Award, 2026.
- University of Iowa Graduate Research Excellence Award Finalist, 2025.
- SC Travel Award, 2024.
- ICDE Travel Award, 2023.

PROFESSIONAL SERVICE

- **Technical Program Committee:** IEEE CLOUD Summit, 2026.
- **Artifact Evaluation Committee:** IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2024.
- **Subreviewer:** DSN 2026, 2023, 2022; HiPC 2025; HPDC 2026, 2023, 2022; ICPP 2026; ICS 2026, 2025; ISSRE 2024, 2023, 2022; Middleware 2022; SELSE 2022; PRDC 2021.
- **Student Volunteer:** SC 2024; ICDE 2023.